

Investigating Geometric Paradoxes

GroupSolve

September 5th, 2026

1 Homework

1.1 Introduction

This week, we investigated three types of Geometric Paradoxes. Only one was relevant enough to dive into, though. Ultimately, it was really just a lesson on Fibonacci Numbers and Continued Fractions. It was a shocking connection, but a strong one indeed. While we explored a few properties, many more properties still linger. The goal of this homework is to explore these topics further.

Feel free to do any of the problems. I really do not care how much you do. As long as you give your best attempt on the problems you attempt. After all, I want you to discover properties! I am just here to guide you, support you, and provide feedback. There really is no minimum amount of problems you have to do. With that being said, there also is no maximum! Feel free to explore continuations of these problems – anything you find – and submit them to me for feedback! In addition, I request only submitting problems you do. Feel free to search online and consult resources, but I humbly request that you only submit solutions to problems you do by yourself.

If you do have questions, which I do expect you will have many as you solve, feel free to contact me. You can also ask me for any tips, advice, and hints! In general, if you want any help with literally anything, including unrelated things to this problem set, please reach out! I promise I am kind and I want to help!

Submit your problems to this link. The due date is 9/19. Good luck!

– Jacob Khohayting

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1.2 Induction

Looking at a few of the sessions I have planned, Induction is going to be a huge part. And so, if there is *anything* you get out of this problem set, I hope it is the idea of induction and how we can use it. We did a few examples in class, but below are few easier examples to get you prepared for some of the later problems.

Recall, Induction is a way of proving something is true for infinitely many things. Some sources would say “we are showing it for $n = 1$, $n = 2$, and so on.” And I will say this is true, but a bit misleading.

In particular, the idea is to show something is true for $n = 1$. Then, you should show that if it is true for some n , then it is true for $n + 1$. It is somewhat misleading as you can do some variation on it. Perhaps you want to show it for $2 \pmod{4}$ numbers, so you show it for $n = 2$, then you can show that if it is true for $n = 4m + 2$ for some integer m , then it is true for $n = 4m + 6$.

It is crucial you understand the idea and you can apply it. If you have any questions, don't hesitate to reach out!

For these problems, I just request you *try* to use induction. However, if you have alternative methods, feel free to show them too!

1. You might know the formulas:

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \text{and} \quad \sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}.$$

Can you prove this by induction?

(If you are unfamiliar with summation notation, this just means the sum of the first n numbers, n squares, and n cubes.)

2. Prove

$$3^0 + 3^1 + 3^2 + \cdots + 3^n = \frac{3^{n+1} - 1}{2}.$$

3. Show that $(1+x)^n \geq 1+nx$ for any real number $x \geq -1$ and any integer $n \geq 0$.

1.3 Fibonacci Sequence

Recall, the *Fibonacci Sequence* is defined by the recurrence $F_0 = 0$, $F_1 = 1$, and for $n > 1$,

$$F_n = F_{n-1} + F_{n-2}.$$

1. Write a list of the first 10 or so Fibonacci Numbers. Sometimes it is good to be familiar with them, just like perfect squares which you may or may not recognize instantly!
2. As n grows large, what value does $\frac{F_n}{F_{n-1}}$ approach?
3. Prove that

$$\sum_{k=1}^n F_k = F_1 + F_2 + \cdots + F_n = F_{n+2} - 1.$$

4. Find closed forms for

- $\sum_{k=1}^n F_{2k},$
- $\sum_{k=1}^n F_{2k-1},$ and

$$\bullet \sum_{k=1}^n F_k^2$$

5. Show every positive integer can be represented uniquely as the sum of one or more distinct, *non-consecutive* Fibonacci numbers.

6. Find the sum of the prime divisors of $12^8 - 1$.

(There is a nice solution; take a peak at all the Fibonacci stuff you've done!)

1.4 Continued Fractions

A simple continued fraction is denoted as

$$[a_0; a_1, a_2, a_3, \dots] = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4 + \frac{1}{a_5 + \ddots}}}}}$$

The n th convergent to a continued fraction $[a_0, a_1, a_2, \dots]$ is the value obtained by cutting off off the fraction after a_n , i.e. the value of $[a_0, a_1, a_2, \dots, a_n]$. In class, we showed that if

$$\frac{p_n}{q_n} = [a_0, a_1, a_2, \dots, a_n],$$

then

$$p_n = p_{n-1} \cdot a_n + p_{n-2}, \text{ and}$$

$$q_n = q_{n-1} \cdot a_n + q_{n-2}.$$

where $p_0 = a_0$, $q_0 = 1$, $p_1 = a_0 a_1 + 1$, and $q_1 = 1$. Furthermore, we showed

$$p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}.$$

1. Find the simple continued fraction of $\frac{2026}{1001}$, $\sqrt{6}$, $\sqrt{10}$, and $\sqrt{14}$. Find the first four "convergents" of each!
2. Find $[1; \overline{1}]$, $[2; \overline{3, 4}]$, and $[5; 6, 7, 8]$. (Note the third example is *non-repeating*.)
3. Show that $\gcd(p_n, q_n) = 1$. In particular, $\frac{p_n}{q_n}$ is a reduced fraction.
4. Show that $p_n q_{n-2} - p_{n-2} q_n = (-1)^n a_n$.

As n goes to ∞ , what do you think $\frac{p_n}{q_n}$ tends toward? Let's see if we can show your hypothesis!

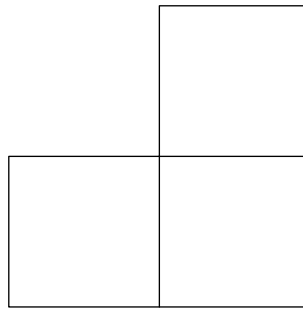
5. Show that the even convergents $\frac{p_{2n}}{q_{2n}}$ is strictly increasing.
6. Show that the odd convergents $\frac{p_{2n+1}}{q_{2n+1}}$ is strictly decreasing.
7. Show every odd convergent is greater than each even convergent.
8. Show that

$$\left| x - \frac{p_n}{q_n} \right| \leq \frac{1}{q_n q_{n+1}}.$$

9. Show an number is rational if and only if it can be expressed as a finite continued fraction.
10. Find an integer solution (x, y) to $1001x + 60y = 1$. (Hint: Try using what we have learned!)
11. Find solutions to $x^2 - 3y^2 = 1$. How about $x^2 - 3y^2 = -2$?

1.5 Ingenuity

1. In a sports tournament of n players, each pair of players plays exactly one match against each other. There are no draws. Prove that the players can be arranged in an order $P_1, P_2, \dots, P_n$ such that P_i defeats P_{i+1} for all i from $1, 2, \dots, n-1$.
2. You have a $2^n \times 2^n$ checkerboard with exactly one square removed. Prove you can cover the rest with L -shaped triominoes without overlap. A L -shaped triomino, which can be rotated, is shown below



3. In a badminton singles tournament, each player played against all the others exactly once and each game had a winner. After all the games, each player listed the names of all the players she defeated as well as the names of all the players defeated by the players defeated by her. For instance, if A defeats B and B defeats C , then in the list of A both B and C are included. Prove that at least one player listed the names of all other players.